



To solve this, note that with $a_1=1$ and $a_2>1$ (from the given code $a_2=1+2\cdot 72^{-10^{24}}+\dots>1$), the optimal schedule is $(x_1, x_2)=(0, 1/a_2)$ (using only the faster drug). Thus the minimiser is approximately $(0, 1)$ (since $1/a_2\approx 1$). Now compare $(0.1, 0.9)$ to $(0, 1/a_2)$: the coordinate differences are $|0.1-0|=0.1$ and $|0.9-1/a_2|\approx |0.9-1|=0.1$. Hence the infinity-norm distance is $\max(0.1, 0.1)=0.1<0.15$. In other words, $(0.1, 0.9)$ lies about 0.1 away from the minimiser, well below the 0.15 threshold.

Answer: Yes – the distance is about 0.10 , which is less than 0.15 , because the true minimiser is essentially $(0, 1)$ and $(0.1, 0.9)$ differs by only 0.1 in each coordinate.
